

Flood Disaster Preparedness in Indian Scenario

Javed Alam¹ and Mohammad Muzzammil²

¹ Reader, Department of Civil Engineering, Zakir Hussain College of Engineering & Technology, Aligarh Muslim University Aligarh, INDIA

Email: javed_alig2000@yahoo.co.in

² Professor, Department of Civil Engineering, Zakir Hussain College of Engineering & Technology, Aligarh Muslim University Aligarh, INDIA

Email: muzzammil786@rediffmail.com

Abstract- Floods are among the most common and destructive natural hazards causing extensive damage to infrastructure, public and private services, the environment, the economy and devastation to human settlements. Recurring flood losses have handicapped the economic development of both developed and developing countries. India is a developing country and prone to a number of natural hazards. Among all the natural disasters that country faces, river floods are the most frequent and often devastating. Through geophysical studies, it has been found that more than one half billion people on our planet reside on river side and coastal flood plains where they produce 1/3 of world's food production and on any day at least some fraction of these plains go under flood water and hence causing wide spread losses to human lives and herds of cattle dead, devastated homes, destroyed the agricultural crops and disrupted the communication links such as railways, roads as well as damage to health hazards. Even after the receding of floods it takes several months or even years before the community comes to the pre-flood level. So floods are a natural and inevitable phenomenon of life in almost all the states of India. Floods depend on many things such as Climate, nature of the Collecting basin, nature of the streams, soil, vegetative cover, amount of snow melt and over all rainfall. India is highly vulnerable to floods and out of the total geographical area of 329 mha, more than 40 mha is flood prone. Floods are recurrent phenomenon, which cause huge loss of lives and damage to livelihood system, property, infrastructure and public utilities. It is a cause of concern that the flood related damages are showing an increasing trend. The average annual flood damage during the last 10 years was about Rs. 5000 crore as compared to Rs. 2000 crore, the corresponding average for the last 50 years. This can be attributed to many reasons including rapid increase in population and urbanisation coupled with growing developmental and economic activities in the flood plains and global warming.

Index Terms- Flood Plains, Global Warming, Precipitation, Natural Hazard

I. INTRODUCTION

The frequency and intensity of floods has grown in the country over the years primarily because of the increased encroachment of flood plains. Interestingly, while the number of deaths caused by flooding has decreased over the last decade, the number of affected populations and economic losses has increased significantly. These trends demand better preparedness at national, provincial and local levels to make

sure that appropriate and effective response measures are taken during flood emergency to minimize the loss of lives and properties. In India, around 75% of the total rainfall is concentrated over 4 months of monsoon (June – September) and, as a result almost all the rivers carry heavy discharge during these four months.. Around 12% of the country's land area is prone to floods which means around 40 million hectares are prone to flood and annually on an average 8 million is affected by floods. The most flood prone are the Brahmaputra, Ganga and the Meghana basins. The states are Uttar Pradesh, Bihar, West Bengal, Assam and Orissa. But of late floods have also become a serious affair in the states of Andhra Pradesh and Gujarat. Over 30 million people are displaced annually.

Recognizing the need of the hour, the decade 1990-99 was declared as "International Decade for Natural Disaster Reduction" with a main objective to focus on disaster management planning for prevention, reduction, mitigation, preparedness and response to reduce the loss of life and property due to natural disasters. India's high risk and vulnerability is highlighted by the fact that 40 million hectares out of a geographical area of 328 million hectares is prone to floods. On an average every year, 75 lakh hectares of land is affected, 1600 lives are lost and the damage caused to crops, houses and public utilities is about Rs.2000 crores due to floods. The maximum number of lives (11,316) was lost in the year 1977. The frequency of major floods is more than once in five years. Floods have also occurred in areas, which were earlier not considered flood prone. Eighty per cent of the precipitation takes place in the monsoon months from June to September. The rivers bring heavy sediment load from the catchments. These, factors coupled with inadequate carrying capacity of the rivers are responsible for causing floods, drainage congestion and erosion of river-banks. The fact is that some of the rivers causing damage in India originate in neighboring countries, adds another complex dimension to the problem. The present paper deals with review on the established practices and the present status of management and preparedness on the floods in India.

II. FLOOD PREPAREDNESS

Floods, which are a natural hazard, need not become a disaster, if we are prepared and are aware of how to deal with them. This would reduce the losses of life and minimize human suffering. This guide list simple thing one can do to stay safe and protect one from floods.

A. Before flooding occurs

- o The route to the nearest safe shelters is to be known.
- o The First Aid Kit is to be ready with extra medication for snake bite and diarrhea.
- o Strong ropes should be available for tying things.
- o A radio, torch and spare batteries are to be arranged.
- o Fresh water, dry food, candles, matchbox, kerosene etc are to be stocked.
- o Umbrellas and bamboo sticks are also necessary to protect from snakes).
- o Higher ground is to be selected for stay where people and animals can take shelter.

B. After hearing a flood warning

- o Flood warning and advice may be easily obtained through radio and television.
- o We must keep vigil of flood warning given by local authorities.
- o Dry food and drinking water and warm clothes are made to be ready.
- o Emergency kit must be checked.

C. At the time of evacuation

- o Pack clothing, essential medication, valuables, personal papers etc in water proof bags to be taken to the safe shelter.
- o Raise furniture, appliances on beds and tables.
- o Put sandbags in the toilet bowl and cover all drain holes to prevent sewage backflow.
- o Do not get into water of unknown depth and current.
- o Lock your house and take the recommended or known evacuation routes for your area of safe shelter.

D. During Floods

- o Boiled water or use of halogen tablet to purify water must be used.
- o Food should be covered.
- o Children are not allowed to remain on empty stomach.
- o Bleaching powder and lime are to be used to disinfect the surroundings.
- o Entry in flood waters may be avoided. If one need to enter then proper foot wear may be used.
- o Water over knee level may be avoided.

E. After a Flood

- o One has to be in touch with local radio.
- o Children may not be allowed to play in, or near, flood waters.
- o One has to be stay away from drains, culverts.
- o Electrical appliances should not be used.
- o Food of floodwaters must be avoided.

- o Tap water should be boiled before use.
- o Halogen tablets must me used before drinking water.
- o One has to be careful of snake bites, snakebites are common during floods.

III. FLOOD PREPAREDNESS PLANNING

Flood preparedness planning is about putting in place a set of appropriate arrangements in advance for an effective response to floods. Some of the commonly identified flood preparedness activities are; Public awareness raising on flood preparedness, response and mitigation measures; Stockpiling of emergency relief materials i.e., food, fodder for livestock, emergency medicines, materials for temporary shelter etc; Installation of community-based early warning system for issuance of timely and effective flood warnings; Management of safe areas for temporary removal of people and property from a threatened location; Transportation to safe areas/ evacuation centre; Ensuring access to health and sanitation facilities; Conducting drills and rehearsals. The key to flood preparedness planning is to have a clarity and agreement on the roles and responsibilities of relevant stakeholders such as the government agencies, disaster management organizations, Red Cross, voluntary groups as well as community members. Such an arrangement is possible by forming disaster management committee and teams at various levels to agree on set of standard operating procedures (SOPs) defining what actions to be taken before, during and after floods. The benefits of the flood preparedness planning are many and some of them are listed:

- o Systematic arrangement and deployment of resources to reduce the impact of flood disaster.
- o Vulnerable communities to get access to crucial information, such as timely flood forecasts and warnings.
- o The provision of basic needs, such as shelter and medical care, clean water, sanitation and food during floods.
- o Continued access to livelihoods, in order to minimize disruption of economic activities; Effective coordination among disaster management agencies to ensure efficient emergency response during floods.
- o Urgent restoration of critical infrastructure and measures to be taken to bring normalcy immediately after the floods.

IV. FLOOD PREPAREDNESS PLAN

A flood preparedness plan (FPP) which is an integral component of the multi-hazard disaster management plan, is an action oriented document detailing specific actions to be undertaken prior to floods, which set the ground for effective execution of emergency response and recovery activities during and after floods. The main components of a FPP are:

- o Assessment of probable needs: Based on historical data from previous flood disasters, officials at the State and district levels compile a list of likely needs and available resources. Gaps between needs and resources are identified in advance and also ways to mobilize them.

- Institutional Mechanism for implementation of FPP: The Flood Preparedness Plan outlines the institutional structure of the States, District or Community level Committees for Disaster Management, its roles and responsibilities before, during and after floods. The Plan also establishes the coordination among the line agencies and other stakeholders in implementation of priority activities identified in the plan.
- Activating early warning and disaster response systems: The FPP defines how to warn the whole community, based on the forecast received from the national and regional agencies and what they should do in advance. The plan ensures ways of involving all stakeholders, according to their roles and responsibilities, and outlines these in the plan.
- Resource mobilization and allocation: Flood control measures require resources; therefore the plan specifies what resources are already available at the State district, community and village levels. The plan also specifies what resources will be needed and where to find those resources.
- Communication within and outside the community: To ensure clear and effective messages in an emergency, the plan specifies how communication will take place and via what media (radio or indigenous systems, etc.)
- Sectoral components: A flood preparedness plan outlines standard operating procedures (SOPs) for specific measures such as search & rescue, emergency medical assistance, provision of water supply and sanitation, food and nutrition, logistics and transport, health, agriculture and environment management, temporary shelter, evacuation procedures; protection and security.

V. FLOOD MANAGEMENT

Though it is not possible to completely avoid the flood disasters, but the sufferings can be minimized by creating proper awareness of the likely floods and its impact by developing a suitable warning system, flood preparedness and management of flood disasters through application of information technology tools. The changing trends have opened up a large number of scientific and technological resources and skills to reduce disaster risk. Providing absolute protection to all flood prone areas for all magnitude of floods of different probabilities of occurrence is neither practically possible nor economically viable. Hence a practical approach in flood management is to provide a reasonable degree of protection against flood damage at economic costs through a combination of structural and non-structural measures.

Structural Measures

The main thrust of flood protection programme undertaken in India so far has been in the nature of structural measures like:

- (i) Embankment, flood walls;(ii) Dams and reservoirs;(iii) Natural detention basin;(iv) Channel improvement;(v) Drainage improvement;(vi) Diversion of flood waters.

Non-structural Measures

Flood plain zoning aims at disseminating information on a wider basis so as to regulate indiscriminate and unplanned development in flood plains and is relevant both for unprotected as well as protected areas. At present the watershed development and management programmes are being implemented under centrally sponsored schemes of Ministry of Agriculture, Ministry of Rural Areas and Employment & Ministry of Forests. Flood proofing measures help greatly in mitigation of distress and provide immediate relief to the population.

VI. FLOOD SIMULATION

Simulation is nothing but a simple mathematical description of the response of a system to a series of events during a selected time period. The basic steps involved in the simulation process are (i) identification of the system; (ii) conceptualization of the system; and (iii) implementation of the model. More complex conceptual models with greater reliability and accuracy have been developed with the widespread use of large, fast electronic computers. National Weather Service River Forecasting System (NWSRFS) has been developed for flood forecasting, navigation, water supply, water control operation and low flow forecasting. Natural phenomena simulation and visualization are becoming very useful in today's information community. Geographic information systems (GIS) are often used for the flood simulation and visualization. Integration of flood simulation model and remote sensing (RS) data with the available topographic and socioeconomic data in a geographic information system (GIS) environment are also used for flood hazard mapping. Hydrologic model in combination with digital elevation model (DEM) is useful in delineating the inundation area extent and estimating the flood depths in areas where images capturing the peak flood events are not available.

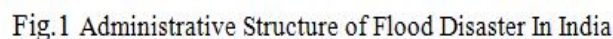
VII. POLICIES ON FLOOD MANAGEMENT

After the unprecedented floods of 1954, the Govt. of India took several Initiatives and constituted a number of committees to study the problems of floods in the country. The important steps are (i) Policy statement 1954 (ii) High level committee on floods – 1957(iii)Policy statement of 1958,Ministerial committee on flood control — 1964 (iv) Ministers committee on floods and flood relief — 1972 (v) Working group on flood control for Five Year Plans, Rashtriya Barh Ayog (RBA) — 1980 (vi) National Water Policy (1987) (vii) National Commission for integrated Water Resource Development Plan 1996 (viii) National Water Policy (Revised in 2002) (ix) National Disaster Management Authority (NDMA 2005) and (x) National Disaster Management Guide Lines (2008).

The activities proposed to be undertaken aim at minimizing the flood risk and losses and are to be implemented in three stages in addition to recurring activities.

These include implementation of the schemes for expansion and modernization of the flood forecasting and

Implementation of activities, which include construction of dams and catchment area treatment (CAT) works in India as well as neighboring countries, is likely to take considerable time as they entail major environmental, social, inter-state and international implications. These need careful study and interaction with the stakeholders. It is envisaged that all feasible schemes will be completed by the year 2025.



IX. PROGRESS OF FLOOD MANAGEMENT

During the last 40 years, different methods of flood protection, both long term and short term, have been adopted in different states depending upon the nature of problems and local conditions. The flood management works carried out, so far, has provided reasonable degree of protection to an area of 14.374 m. ha up to March 1993. and raising of 4,721 villages above flood levels

The works consists of 16,199 km of new embankments, 32,003 km of drainage channels, 906 town protection works Reservoirs constructed on the Damodar river, across Mahanadi (at Hirakund), Sutlej (at Bhakra), Brahmaputra (at Rengali) and Tapi, (at Ukai), as also on a number of other major and medium reservoirs, have helped greatly in reducing the intensity of flood in the flood plains. Another important non-structural measure for mitigating damages from floods is Flood Plain Zoning, out of 1, 06,000 sq.km, programmed for surveys in the country, about 55,000 sq.km of flood prone area was surveyed up to March 1991, by the Survey of India, for preparation of large scale maps of 1:15000 with contour intervals of 0.3 to 0.5 m. The maps to be used by the State Governments to demarcate flood plains and to prepare Flood Risk Maps for introducing suitable regulatory measures in the flood plains. All the works have resulted in providing reasonable degree of protection to total area of about 14 m ha against the total flood prone area of 40 m. ha in the country. Total investment on flood control during the last four decades has been of the order of about Rs. 40 billion.

X. FINANCIAL ALLOCATION FOR FLOOD

Flood Disasters are huge economic burdens on developing economies such as India. Every year huge amount of resources are mobilized for rescue, relief and rehabilitation works following flood disaster occurrences. The Central Government plays a major role as far as mobilization of financial resources is concerned. A scheme called Calamity Relief Fund (CRF) has been constituted for each state with contribution from the Central and State Government to undertake relief and rehabilitation measures. This enable the states to manage and provide for calamity relief on their own by drawing upon the resources available with a fund constituted for that purpose separately for each state. In addition to CRF, a National Fund for Calamity Relief (NFCR) has been created to deal with hazards of rare severity managed by a National Calamity Relief Committee (NCRC). The State Governments are required to submit memoranda for this purpose giving details of damage and destruction and the cost of relief and rehabilitation. On receipt of these memoranda, the Government of India decides on an individual basis whether a Central Team is required to be deputed to assess the situation. The administrative structure of Flood Disaster in India is shown in Fig.1.

X.I FLOOD RELIEF ARRANGEMENTS

If efforts of the civil authorities for organizing rescue operations and shelter are considered inadequate, Army assistance should be requisitioned. Relief measures by non-official and voluntary organizations may be enlisted as far as possible. Relief camps should be organized. Provision of basic amenities like drinking water, sanitation and public health care and arrangements of cooked food are to be made in the relief camps. Necessary arrangements for air dropping of food packets in the marooned villages through helicopters must be made. Enough relief parties to the rescue of the marooned people within a reasonable time limit are to be organized. Alternate and effective communication links for communication with marooned areas are to be established. Controlled kitchens for food supply must be organized initially at least for 3 days. Cattle camps are to be organized, if necessary, and veterinary care, fodder and cattle feed to the affected animals are to be provided. Grant of emergency relief to all the affected people must be assured.. Submission of daily reports and dissemination of correct information through mass media are to be made to avoid rumors. When floods are over; some important measures would have to be taken: (i) Rehabilitation of homeless. (ii) Commencement of agricultural activities. (iii) Re-sowing. (iv) Repair and reconstruction of infrastructural facilities such as roads and embankments. (v) Resettlement of flood prone areas. (vi) Health measures. (vii) Relief for economic reconstruction.

XII. COMMUNITY PARTICIPATION

Flood being a common concern for all sections of society living in the affected area, flood management needs active involvement and participation of all to fulfill its objective. The national Water Policy also provides for involvement of beneficiaries in various aspects of management of Irrigation & Flood Control Schemes. If people, living in the flood plains, are fully conversant with the system of warning signals, issued from time to time in different situations, messages are disseminated to the people living in affected areas quickly and they are fully involved, the loss of life and property can be minimized to a considerable extent.

XIII. CONCLUSIONS

‘Forewarned is forearmed’, goes the saying, and this country, the worlds second most populous and with a large proportion of its people desperately poor, must prepare as best as it can to cope with this new trend of heavy rains and floods. Several steps are, in fact, being taken. The Indian Meteorological Department (IMD) has been promised a major up gradation of its observational network, which should help improve the ability to forecast episodes of unusually heavy rain at least a day or two in advance. The Indian Space Research Organization is creating a digital database with detailed terrain information that can be used for hazard zonation and damage assessment. The space agency is also acquiring an aircraft that will be equipped with airborne radar so that imageries of disaster-hit areas can be quickly acquired

and used to supplement information from earth-viewing satellites. A National Disaster Management Authority has been established and similar bodies are to be set up at the State level too to enhance preparedness and provide a coordinated response in the event of a calamity. But still it is most important to make sure that all these measures come together properly in order to improve resilience and enable an effective relief operation when catastrophes strike. More active participation is needed from voluntary organizations in case of a disaster. More emphasis should be given on training of people to cope with the flood situations. Proper post flood measures should be incorporated in the NDMA guide lines.

REFERENCES

- [1] Sinha, Anil & Sharma, K. Vinod, "Culture of Prevention, Government of India", Ministry of Agriculture, Natural Disaster Management Division, New Delhi(1999),.
- [2] G. S. Mandal, "Forecasting and Warning Systems for Cyclones in India", Shelter, pp. 24-26. October, 1999
- [3] A. Sinha, "Relief Administration and Capacity Building for Coping Mechanism towards Disaster Reduction", Shelter, pp. 9-12, October 1999.
- [4] R. Singh," Practices and current status of flood forecasting and Management in India", Emeritus Fellow, WRDTC, Indian Institute of Technology, Roorkee, India.
- [5] Flood Preparedness, a necessity: an article in The Hindu (online edition) Sept 19th 2007.
- [6] Flood Preparedness Planning by Aslam Perwaiz , ADPC – Asian Disaster Management News Vol 12 No 3
- [7] S. P.Meyer, T. H. Salem, and J. W. Labadie, "Geographical Information System in Urban Storm-Water Management", Journal of Water Resources Planning and Management, Vol. 119, No.2. 1993.
- [8] F. Giorgi, "Simulation of regional climate using a limited area model nested in a general circulation model", J. Climate, 3, 941-963, 1990.
- [9] M. R. Leipnik, K. K. Kemp, and H. A. Loaiciga, "Implementation of GIS for Water Resources Planning and Management", Journal of Water Resources Planning and Management, Vol. 119, No.2. 1993.
- [10] Y.-Q. Lui, F. Giorgi, and W.Mm Washington, "Simulation of summer monsoon climate over East Asia with an NCAR regional climate model", Mon. Wea. Rev., 122, 2331-2348, 1994.
- [11] D.Parker and M. Fordham, "An Evaluation of Flood Forecasting, Warning and response System in the European Union", Vol. 10, 1996.
- [12] P. A. Townsend and S. J. Walsh, "Modelling Flood Plain Inundation using Integrated GIS with Radar and Optical Remote Sensing", *Geomorphology* 21 (98), pp. 295-312, 1998
- [13] M. M. Islam and K.Sadu, "Flood Damage and Modelling using Satellite Remote Sensing Data with GIS: Case Study of Bangladesh", 2001.
- [14] Hu Heping, Guo Jianzhong, and Weng Wenbin, "Theory and application of Urban Flood Dynamic Simulation Model, Proceedings of the Basic Research on City and Engineering Disaster Mitigation", Guangzhou, China, 1997.
- [15] Md. slam Monirul and Sado Kimiteru, "Development Priority Map for Flood Countermeasures by Remote Sensing Data with Geographic Information System", *Journal of Hydrologic Engineering/September/October 2002*, pp. 346-355, 2002.
- [16] Sanyal Joy and Lu Xi Xi, "Application of Remote Sensing in Flood Management with Special Reference to Monsoon Asia – A Review", *Natural Hazards* 33, Kluwer Academic Publishers, 2004.
- [17] Sanyal Joy and Lu Xi, "Remote Sensing and GIS-Based Flood Vulnerability Assessment of Human Settlements: A Case study of Gangetic West Bengal, India", *Hydrological Processes* 19, pp. 3699-3716, 2005
- [18] S.YAMAGUCHI, and K. WAMURA, "Fast Flood Simulation Method using Dynamic DDM", *IPSJ Transactions on Mathematical Modeling and Applications*, 48(6), 92-103 (in Japanese).2007
- [19] Second Irrigation Commission of India 1972.
- [20] Reports of Rashtriya Barh Ayog (RBA) — 1980.
- [21] Water Vision 2050 Indian Water Resources Society, (IWRS) 1999.
- [22] Five decades of Water Resources Development of India (IWRS) —1998.